Vector Spaces and Subspaces

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Vector spaces generalize coordinate vectors. Subspaces are closed under linear combinations; column and null spaces record the action of a matrix, while bases and dimension measure independent directions.

Spaces and Subspaces

Subspace test:
$S \leq V$ iff $0 \in S$ and $u, v \in S$, $c, d \in \mathbb{R}$ imply $cu + dv \in S$.

$$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\} = \text{span}\{a_1, \ldots, a_n\}, \quad \mathcal{N}(A) = \{x : Ax = 0\}.$$

$$\mathcal{R}(A) = \text{row}(A), \qquad \mathcal{N}(A^T) = \text{left nullspace}(A).$$

Basis theorem:
a basis is an independent spanning set. Every vector has a unique coordinate representation $v = \sum c_i q_i$.

$$\dim S = \text{number of vectors in any basis of } S.$$

Dimension theorem:
$\dim \mathcal{C}(A) = \text{rank}(A) = \dim \mathcal{R}(A)$.

Rank and Nullity

Rank–nullity theorem:
for $A : m \times n$, $\dim \mathcal{N}(A) + \text{rank}(A) = n$.

$$\dim \mathcal{C}(A) + \dim \mathcal{N}(A^T) = m.$$

Four-subspace dimension relations:
$\dim \mathcal{R}(A) = \dim \mathcal{C}(A) = r$, $\dim \mathcal{N}(A) = n - r$, $\dim \mathcal{N}(A^T) = m - r$.

Fundamental theorem:
$\mathcal{R}(A) \perp \mathcal{N}(A)$ and $\mathcal{C}(A) \perp \mathcal{N}(A^T)$.

$$\text{rank}(AB) \leq \min(\text{rank}\, A, \text{rank}\, B), \quad \text{rank}(A + B) \leq \text{rank}\, A + \text{rank}\, B.$$